

Aaryav Chaudhary

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Education

Indian Institute of Information Technology (IIIT), Bhopal

Sep 2023 – Jul 2027

Bachelor of Technology in Computer Science and Engineering

- Coursework: Machine Learning, Deep Learning, Natural Language Processing (NLP), Data Structures and Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS)

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL

AI/ML: Large Language Models (LLMs), Natural Language Processing (NLP), Transformers, Fine-tuning, Prompt Engineering, Retrieval-Augmented Generation (RAG)

Frameworks and Tools: PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, FAISS, OpenAI API, Groq API, Google Gemini API, Scikit-learn, Streamlit

Backend and Infrastructure: FastAPI, Node.js, Express.js, REST APIs, Redis, Docker, Git, AWS, PostgreSQL, MongoDB, Continuous Integration/Continuous Deployment (CI/CD)

Experience

University of Galway, Ireland

Sep 2025 – Mar 2026

Research Intern, Federated Learning and RAG

Remote

- Engineered privacy-preserving Federated Learning techniques for LLMs using differential privacy in Python, reaching 95% of centralized accuracy with 30% stronger regulatory compliance across distributed nodes.
- Developed hybrid RAG pipelines (BM25 + dense embeddings) with FAISS vector indexing, improving retrieval efficiency by 40% and cutting inference latency by 25%.

Projects

RecruitmentAgent – Multi-Agent Career Platform | Python, LangChain, Groq, FAISS, LiveKit

GitHub | Live Demo

- Architected a multi-agent recruitment platform using LangChain and Groq LLaMA-3, combining RAG-based resume parsing, JD-vs-resume ATS scoring, and automated LaTeX resume generation with FAISS vector search.
- Designed a 3-tier job search pipeline (JobSpy, SerpAPI, fallback) across LinkedIn, Indeed, and Glassdoor, plus a real-time AI-avatar interview agent powered by LiveKit and Groq STT/TTS.

HealthGPT – Medical QA System | Python, FastAPI, LangGraph, FAISS, Llama 3.2

GitHub | Live Demo

- Developed a Medical Question Answering system using LangGraph DAG orchestration with 2-stage hybrid retrieval (FAISS + BM25 with Reciprocal Rank Fusion) and Cross-Encoder reranking, achieving 90%+ answer faithfulness on RAGAS evaluation metrics.
- Deployed containerized backend via Docker Compose with Prometheus monitoring, CI/CD via GitHub Actions, and Server-Sent Events streaming for real-time token delivery at 3–8 tokens/sec on CPU-only hardware.

AI Data Analyst – Conversational Analytics Platform | Python, FastAPI, React, Google Gemini, Plotly

GitHub | Live Demo

- Engineered a full-stack AI data analysis platform with FastAPI and React, letting users query CSV datasets in natural language through a 6-tool LLM router built on Google Gemini.
- Implemented automated chart generation, IQR/Z-score anomaly detection, and SQL/Pandas code generation, streaming responses in real time via Server-Sent Events for fast, multi-file data exploration.

Yatra-Vritta – AI Travel Engine | Python, Flask, Scikit-learn, Node.js, MongoDB, AWS

GitHub | Live Demo

- Built an AI travel recommendation engine using Python with Scikit-learn, implementing a K-Nearest Neighbors ML model achieving 92% match accuracy, handling 1,000+ concurrent requests/day at sub-200ms latency on AWS.
- Orchestrated event-driven microservices with Flask REST API and MongoDB, implementing HIPAA and GDPR-compliant data handling and reducing trip-planning time by 45%.

Certifications and Achievements

DeepLearning.AI (Coursera): Machine Learning, Deep Learning, and Generative AI with LLMs by Andrew Ng.

Amazon ML Challenge: Ranked top 200 out of 75,000+ teams in Amazon's national Machine Learning competition.

LeetCode: Solved 500+ Data Structures and Algorithms problems.